

NB: No lymphatics in the CNS.

Brain Barriers

A. Blood Brain Barrier:

Barrier between blood and nerve cells.

Prevents the passage of some substances, such as antibiotics, chemicals and bacterial toxic matter, from blood to the nerve tissue.

Structure:

- Tight junctions between non fenestrated endothelium lining capillaries.
- Thick and continuous BM.
- Astrocyte End-Feet: outside BM.

B. Blood CSF Barrier:

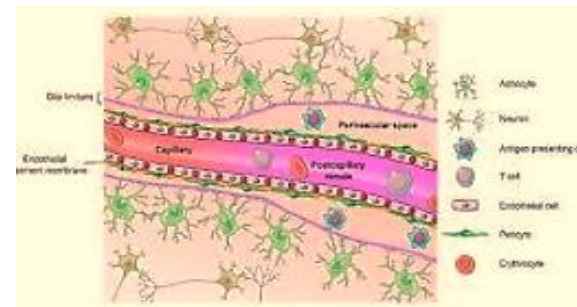
Separates blood from CSF.

Structure:

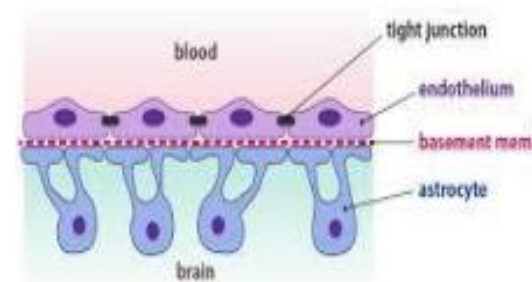
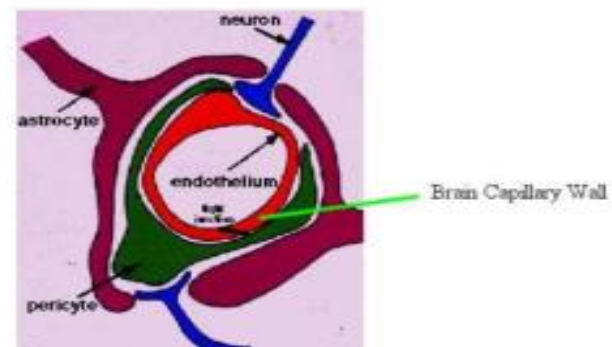
- Tight junctions between fenestrated endothelium lining capillaries.
- BM of fenestrated endothelium.
- BM of ependymal cells
- Ependymal cells.

C. Brain CSF Barrier:

- Ependymal cells.
- Basement membrane.
- Astrocyte End-Feet



Blood brain barrier



Cerebral & Cerebellar cortex

A. Brain:

1. **Cerebrum:** composed of 2 cerebral hemispheres.
2. **Cerebellum:** composed of 2 cerebellar hemispheres.

3. Brain stem

B. Spinal cord.

Gray matter: Composed of nerve cell bodies, dendrites, the initial unmyelinated nerve axon, glial cells, and synapses.

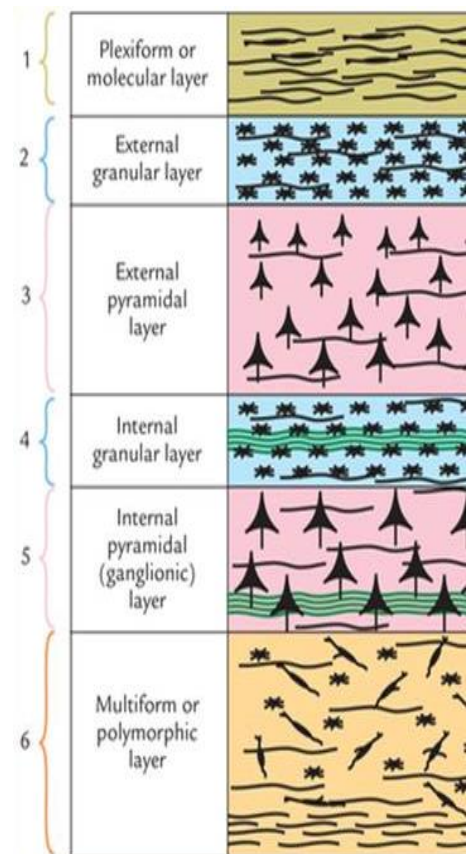
White matter: Composed mainly of the myelinated axons together with the myelin producing oligodendrocytes.

1. Cerebral cortex

The cerebral cortex is organized into six horizontal layers. From superficial to deep, the six layers are:

1. **Molecular layer:** fiber layer; apical dendrites and nerve terminals originating from deeper layers.
2. **Outer granular layer:** formed of granule (stellate) cells and neuroglial cells.
3. **Outer pyramidal layer:** small and medium sized pyramidal cells
4. **Inner granular layer:** formed of small granular cells and neuroglial cells.
5. **Inner pyramidal layer:** large pyramidal cells
6. **Multiform layer:** variably shaped cells.

A sensory region (its neurons integrate sensory information) has a thick internal granule layer. In the primary motor cortex of the frontal lobe, inner pyramidal layer contains giant pyramidal cells called Betz cells, whose axons travel through the internal capsule, the brain stem, and the spinal cord forming the corticospinal tract, which is the main pathway for voluntary motor control.

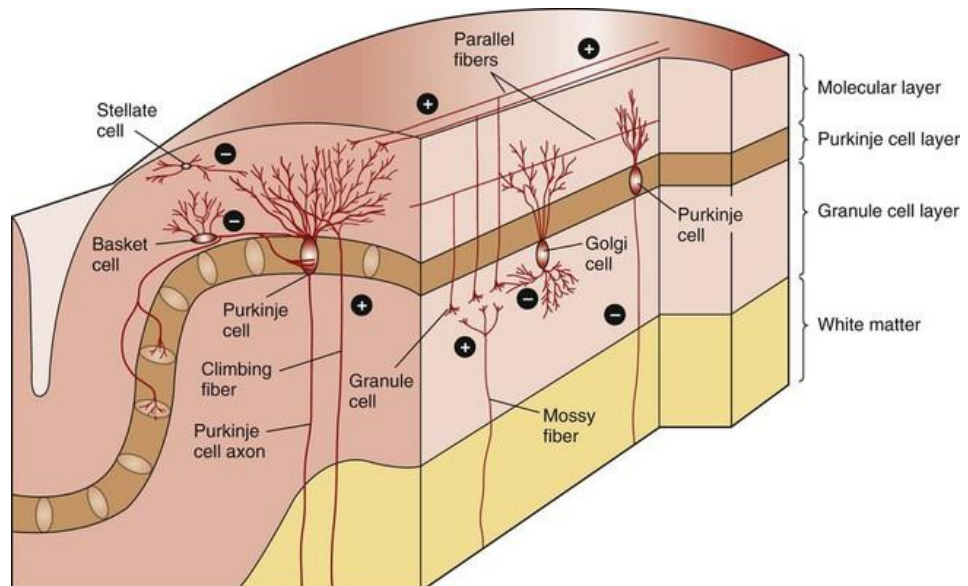


2. Cerebellar cortex

The cerebellar cortex coordinates muscular activity throughout the body.

It has three layers:

1. **Molecular layer:** Outermost layer of grey matter, containing dendritic tree of purkinje fibers, T shaped axons from the inner granular layer, stellate cells and basket cells that have dendritic processes that can contact Purkinje cells dendrites.
2. **Purkinje cell layer:** huge neurons that have fine dendrites and axons. Purkinje dendrite extends as fan or tree into the molecular layer (outer layer), while their axons extend into the granular layer then to the white matter.
3. **Granular layer:** The innermost layer, containing granule cells. Their T shaped axons that extend into the molecular layer to meet up with the Purkinje dendrites. Golgi cells are also scattered in this layer.



Nerve fiber myelination

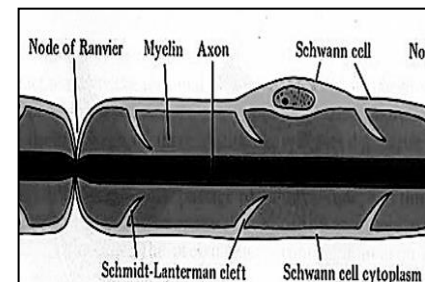
Myelin is a fatty white substance that surrounds the axon of some nerve cells forming an electrically insulating layer formed by Schwann cells in PNS- Oligodendrocytes in CNS. Myelinated axons are white in appearance, hence the "white matter" of the brain.

Functions of myelin:

1. Myelin helps to protect and insulate the axons from electrically charged atoms and molecules in the fluid surrounding the entire nervous system, where the impulse travels by hopping.
2. Rapid conduction of impulses
3. Myelination also prevents the electric current from leaving the axon.
4. When a peripheral fiber is injured, the myelin sheath provides a track along which regrowth can occur.

Areas with no myelin:

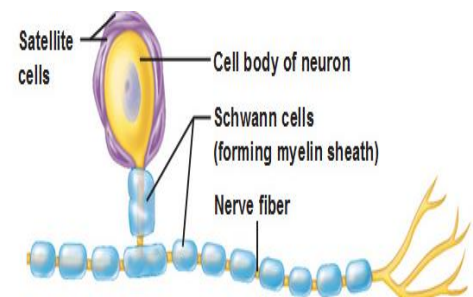
- Axon hillock
- Nodes of Ranvier: the gap between 2 Schwann cells.
- Schmidt Lantermann clefts: oblique interruptions of myelin sheath because of Schwann cell cytoplasm trapped within the lamellae of myelin.
- Axon terminal



Schwann Cells or neurolemma cells:

Each Schwann cell wraps its membrane (lipids and proteins) around the axons in concentric circles to form myelin (the white shiny fatty substance). This squeezes out all the cytoplasm from that region, leaving only the lipid-bilayer, pushing the cytoplasm and nucleus to the periphery. This greatly improves electrical conduction in the axon. Many Schwann cells are needed to cover the whole length of a single axon.

In certain diseases **as multiple sclerosis**, the myelin sheaths in the CNS are destroyed, and action potentials are slowed.



Types of nerve fibers:

1. **Naked:** not covered by myelin or neurolemma.
2. **Myelinated without neurolemma:** nerves in white matter in CNS.
3. **Myelinated with neurolemma:** in PNS
4. **Non myelinated with neurolemma:** the nerve is present in deep invagination in the Schwann cell cytoplasm: in autonomic nerves.

Nodes of Ranvier: There are unmyelinated gaps between myelin sheaths surrounding an axon that we call nodes of Ranvier. (The nodes of Ranvier, being exposed to the ECF every 1 mm or so, contain large amounts of voltage-gated sodium channels, and allow enough sodium into the axon to regenerate the action potential .

Axonal transport:

1. **Anterograde transport:** conducting electrical impulses, proteins, lipids, and other macromolecules, from the cell body to the distal end. There are 2 types:
 - **Slow anterograde:** transport system of a few millimeters a day.
 - **Fast anterograde:** about 100 times faster, involves transport via microtubules, and the microtubule motor protein.
2. **Retrograde transport:** recycled constituents are returned to the cell body.

1. Nerve Endings (Receptors)

2. Free nerve endings:

The most common nerve endings in skin.

Structure: Unencapsulated receptors where the nerve loses its myelin.

Functions: Free nerve endings sensitive to pain, hot and to light touch

3. Merkel's disc:

Structure: Unencapsulated receptors.

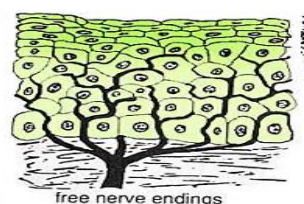
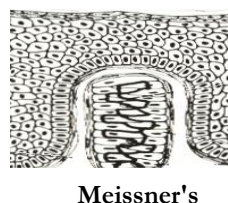
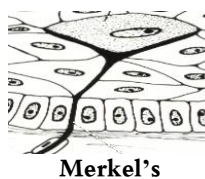
Site: Merkel's disks are near the base of the epidermis.

Function: light touch (discriminative touch).

4. Meissner's corpuscles:

Structure: Surrounded by CT capsule

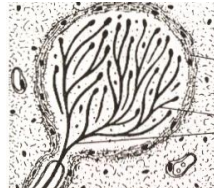
Function: in the dermal papillae, responsible for touch sensitivity (tactile).



5. Krause end bulb:

Structure: Encapsulated receptor.

Function: they are mechanoreceptors and thermoreceptors that respond to cold weather.



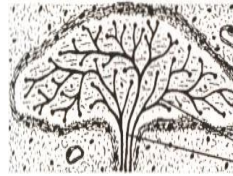
Krause end

6. Ruffini endings:

Structure: Surrounded by a CT capsule.

Site: located deeper in the dermis.

Function: mechanoreceptors for skin stretch and deformations within joints. also detect warm weather.

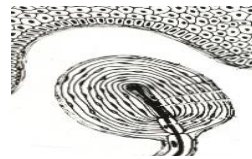


Ruffini corpuscle

7. Pacinian corpuscles:

They are surrounded by a strong CT capsule.

Function: pressure-sensitive mechanoreceptors, present in joint and pancreas.



Pacinian corpuscle

Muscle spindle

Muscle spindles are stretch receptors that signal the length and changes in length of muscles. They lie within an independent capsule, parallel to the main muscle. They are therefore stretched when the muscle lengthens but shorten when the muscle contracts.

Structure:

Skeletal muscle fibers

Specialized sensory organs (proprioceptors).

Have a collagenous CT capsule.

Capsule is fusiform shape so called (intrafusal).

Composed of 3-12 intrafusal muscle fibers.

Types:

1. **Nuclear bag:**(1-2 fibers), have a clustering of nuclei in middle of the spindle.
2. **Nuclear chain fibers:** (5-7 fibers) the nuclei are spread out like a chain.

Innervation:

Motor supply to muscle spindle: Controls the muscle tone.

1. **Extrafusal fibers** are innervated from CNS by alpha motor neurons.
2. **Intrafusal Fibers** are innervated from CNS by gamma motor neurons.

Sensory Endings in Muscles: send information to CNS about proprioception (sensing Muscle Condition and Limb Position). They are of 2 types:

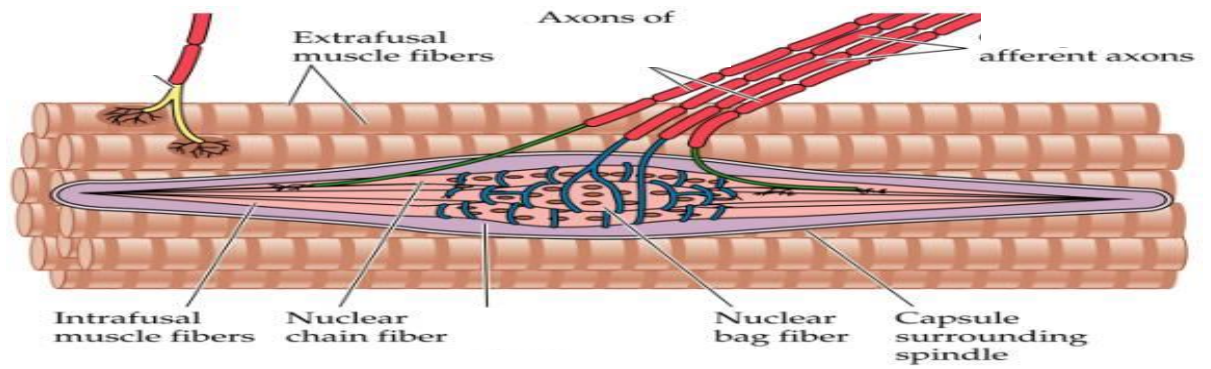
1. **Anulo-spiral** wraps around muscle spindle, senses dynamic changes in

Neuro 25-26

muscle length.

2. **Flower-spray** looks like little flowers, sense static changes in muscle length, helps determine position.

Function: send informations about muscle stretch to CNS.



Muscle spindle